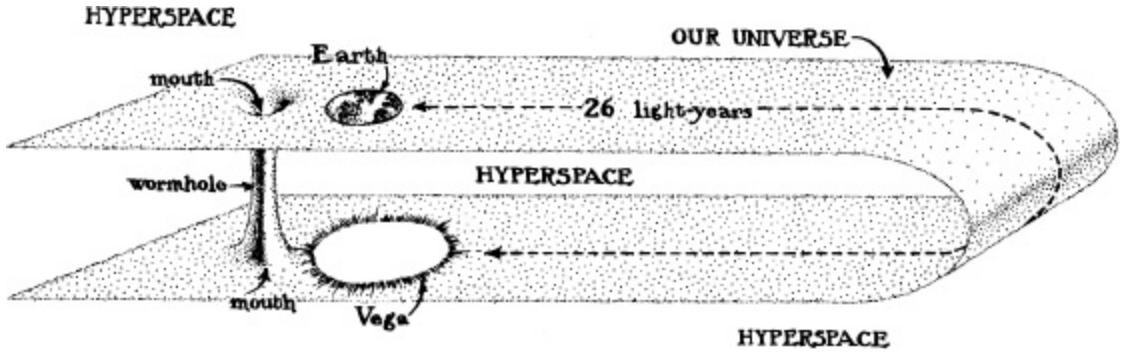
folded, as in the diagram. However, the gentle fold is important; it permits the Earth and Vega to be near each other in hyperspace so they can be connected by the short wormhole. With the wormhole in place, we, like an ant or worm crawling over the embedding diagram's surface, have two possible routes from Earth to Vega: the long, 26-light-year route through the external Universe, and the short, 1-kilometer route through the wormhole.



14.1 A1-kilometer-Iong wormhole through hyperspace linking the Earth to the neighborhood of Vega, 26 light-years away. (Not drawn to scale.)

What would the wormhole's mouth look like, if it were on Earth, in front of us? In the diagram's two-dimensional universe the wormhole's mouth is drawn as a circle; therefore, in our three-dimensional Universe it would be the three-dimensional analogue of a circle; it would be a sphere. In fact, the mouth would look something like the spherical horizon of a nonrotating black hole, with one key exception: The horizon is a "one-way" surface; anything can go in, but nothing can come out. By contrast, the wormhole mouth is a "two-way" surface; we can cross it in both directions, inward into the wormhole, and back outward to the external Universe. Looking into the spherical mouth, we can see light from Vega; the light has entered the other mouth near Vega and has traveled through the wormhole, as though the wormhole were a light pipe or optical fiber, to the near-Earth mouth, where it now emerges and strikes us in the eyes.

Wormholes are not mere figments of a science fiction writer's imagination. They were discovered by Ludwig Flamm mathematically, as a solution to Einstein's field equation, in 1916, just a few months after Einstein formulated his equation; Einstein and Nathan Rosen explored them in the 1930s; and John Wheeler and his research group studied them extensively, by a variety of mathematical calculations, in the 1950s. However, none of the wormholes that had been found as solutions of